

Repair 1 - Description and distributions

Scope: STAT1-STAT2

- Show center plus spread and shape.
- Outlier rule flags; it does not command deletion.
- Compare mean/median before normal-model language.
- z-score = relative distance, not an error detector.

Repair discipline

- Keep first attempt/evidence.
- Repair the conceptual error, not just the formula cell.
- Close review before changed retry.
- Recheck interpretation language after arithmetic is fixed.

Repair 2 - Sampling, SE and CI

Scope: STAT3-STAT5

- Write target population, frame and who can never be selected.
- Large n improves precision but does not repair selection bias.
- Do not confuse raw-data SD with estimator SE.
- CI width is precision; narrow does not prove unbiased.

Repair discipline

- Keep first attempt/evidence.
- Repair the conceptual error, not just the formula cell.
- Close review before changed retry.
- Recheck interpretation language after arithmetic is fixed.

Repair 3 - p-values and effect size

Scope: STAT6-STAT7

- p-value is conditional on H_0 , not $P(H_0 \text{ true})$.
- Fail to reject is not proof of equality/no effect.
- Report absolute pp and relative % effect separately.
- Compare effect with predeclared worthwhile threshold and guardrail.

Repair discipline

- Keep first attempt/evidence.
- Repair the conceptual error, not just the formula cell.
- Close review before changed retry.
- Recheck interpretation language after arithmetic is fixed.

Repair 4 - Correlation and causality

Scope: **STAT8**

- Inspect scatter/outliers before quoting r .
- Name confounding, reverse causality and selection alternatives.
- Observational correlation supports association wording only.
- Propose randomized/stronger design for causal identification.

Repair discipline

- Keep first attempt/evidence.
- Repair the conceptual error, not just the formula cell.
- Close review before changed retry.
- Recheck interpretation language after arithmetic is fixed.

Repair 5 - A/B design and Gate

Scope: STAT9 / M-STAT / Gate

- Predeclare unit/metric/guardrail/duration/threshold before results.
- Check group counts/IDs/labels before statistics.
- Decision uses design + effect + uncertainty/p + threshold + guardrail.
- Gate B/C require prior failure + saved attempt + review + repair + explicit assignment; no automatic progression.

Repair discipline

- Keep first attempt/evidence.
- Repair the conceptual error, not just the formula cell.
- Close review before changed retry.
- Recheck interpretation language after arithmetic is fixed.